# TacBFP: A Tactile-conditioned Behavior Foundation Prior Model for Dexterous Reorientation

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***Abstract*—Dexterous in-hand manipulation requires policies that coordinate many coupled hand joints through intermittent, contact-rich interaction. Current reinforcement learning pipelines can train strong specialist policies for a particular object, pose distribution, or sensing configuration, but each new downstream task often requires expensive exploration from low-level joint actions. We propose TacBFP, a tactile behavior foundation model for dexterous robotic hands. The central idea is to distill reusable contact behavior from a controlled family of simple reorientation skills: we first train eight sphere reorientation specialists over scale factors from $0.3$ to $1.0$, and then distill their closed-loop behavior into a single generalist variational latent controller with a learned tactile-conditioned prior and decoder. The learned prior is conditioned on proprioceptive and tactile history, including per-finger contact magnitudes, contact locations, and object scale cues, so latent codes remain grounded in the current hand-object interaction state. Downstream policies keep the prior and decoder frozen and learn only a residual latent action, enabling adaptation to harder anisotropic objects such as ducks and strawberries while preserving stable contact behavior. We evaluate TacBFP as a reusable motor substrate for complex-object reorientation, arm-hand manipulation, and downstream learning settings.**

***Index Terms*—Dexterous manipulation, behavior priors, tactile sensing, policy distillation, reinforcement learning.**

## I. Introduction

DEXTEROUS robotic hands promise general-purpose manipulation because they can rotate, stabilize, and regrasp objects without relying on external fixtures. This capability is important for service robots, industrial manipulation, assistive systems, and teleoperated platforms, where objects vary in shape, scale, pose, friction, and sensing availability. However, dexterous control remains difficult because useful behavior is not only a sequence of joint targets: it is a contact strategy that must react to the object's motion, the hand configuration, and tactile feedback at the fingertips.

Reinforcement learning has made substantial progress on dexterous manipulation by training specialist policies in large-scale simulation [1], [2]. These policies can solve challenging reorientation tasks when trained with carefully designed rewards, domain randomization, and privileged object information. Yet this paradigm scales poorly as a reusable manipulation system. A specialist policy trained for one object family, target distribution, or observation setting is usually not a convenient starting point for another task, because downstream learning must again explore in a high-dimensional low-level action space. This is especially costly for dexterous hands, where random joint actions frequently break contact, drop the object, or enter unstable configurations.

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A promising alternative is to learn a reusable behavior representation and expose only a compact latent action to downstream tasks. Prior work in physics-based control has shown that broad motor skills can be distilled into an encoder-decoder model with a learned prior, and that downstream reinforcement learning can act through residual latent codes instead of raw joint targets [3]. The key insight is that a learned prior can guide exploration toward coherent motion before task-specific rewards provide dense supervision. Dexterous manipulation has a similar need, but the representation must be contact-aware: a useful hand prior should change its latent distribution when a fingertip gains or loses contact, when contact force shifts across fingers, or when the object begins to slip.

In this paper, we propose TacBFP, a tactile behavior foundation model for dexterous in-hand reorientation. Rather than collecting demonstrations from many complex objects directly, we first train eight specialist policies for a sphere with scale factors from 0.3 to 1.0. This controlled teacher family exposes the hand to different contact apertures, contact-force patterns, and rolling/regrasping regimes while keeping object geometry simple. We then distill these specialists into a single generalist variational latent controller consisting of an encoder $q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{g})$, a learned prior $p_\theta(\mathbf{z}|\mathbf{x})$, and a decoder $\pi_\psi(\mathbf{a}|\mathbf{x}, \mathbf{z})$. Unlike a purely proprioceptive motion prior, TacBFP conditions $\mathbf{x}$ on a short history of hand joint states, previous control targets, tactile contact magnitudes, tactile contact positions, and object scale cues. As a result, the prior and decoder are explicitly grounded in the current tactile interaction state.

After distillation, the prior and decoder are frozen and reused by downstream policies. A task policy predicts a residual latent action $\Delta\mathbf{z}$ from task observations, and the executed latent code is $\mathbf{z} = \mu_{\text{prior}} + \Delta\mathbf{z}$. The decoder converts this latent code into the 22-dimensional hand action. This design preserves the stable contact behaviors learned from specialists while allowing downstream tasks to adjust intent through a low-dimensional residual. It also makes the same behavior model compatible with different high-level inputs, including object goals, visual observations, real-robot observations, or demonstrations.

Our contributions are:

- We formulate dexterous behavior foundation modeling as specialist-to-generalist distillation from multi-scale sphere reorientation policies.
- We design a tactile-conditioned prior and decoder with a direct-concatenation MLP state encoder, allowing the latent distribution to adapt to contact force, contact position, and object scale.

- We reuse the frozen prior and decoder for downstream reinforcement learning through residual latent control, reducing the downstream policy's action space from low-level hand commands to a compact latent correction.
- We evaluate the learned prior on harder anisotropic objects and arm-hand manipulation, testing whether contact behavior learned from simple spheres transfers to more complex downstream settings.

## II. Related Work

### A. Reusable Skill Representations for Physics-Based Control

Reusable latent spaces have been widely studied for physics-based control. Adversarial skill embeddings such as ASE and CALM learn compact skill codes that map random or user-conditioned latents to realistic character behavior [4], [5]. Explicit imitation-based approaches, including NPMP, ControlVAE, PhysicsVAE, and neural categorical priors, learn latent motor representations through reconstruction, dynamics modeling, or imitation objectives [6]–[9]. Recent humanoid behavior priors further show that broad motor skills can be distilled into variational latent spaces with learned proprioceptive priors, enabling downstream tasks to sample coherent whole-body behavior [3]. TacBFP studies a different control regime: contact-rich dexterous manipulation, where the reusable behavior model must be conditioned on tactile contact history rather than only body kinematics.

### B. Dexterous Manipulation with Reinforcement Learning

Deep reinforcement learning has enabled simulated dexterous hands to solve object rotation, reorientation, and manipulation tasks under randomized dynamics. Early demonstration-augmented policy gradients showed that demonstrations can make high-dimensional dexterous manipulation tractable [10]. Large-scale model-free systems then achieved broad in-hand object reorientation over thousands of shapes and distilled policies toward deployable observations [11]. More recent systems improve generalization and sim-to-real transfer through rapid motor adaptation and multimodal sensing [12], [13]. These systems demonstrate strong task performance, but the resulting policies are usually trained for a particular objective and observation interface. TacBFP instead distills specialist behavior into a reusable tactile-conditioned latent prior that can be frozen and reused by multiple downstream policies.

### C. Tactile Sensing for Contact-Rich Control

Tactile sensing provides direct evidence about contact state, load distribution, and incipient slip. Touch-only and visuotactile in-hand manipulation systems show that tactile signals can support object rotation, improve state inference under occlusion, and reduce dependence on visual object tracking [13]–[15]. Recent tactile representation learning and visuotactile imitation learning further show that pretrained tactile encoders and unified 3D visuotactile representations can improve fine-grained manipulation and long-horizon contact-rich tasks [16], [17]. In our implementation, the student-visible state contains smoothed contact magnitudes for five tactile sites and 3D contact positions for each site over a short temporal window. This differs from latent priors conditioned only on kinematics: the distribution over latent actions can change immediately when the contact pattern changes.

### D. Policy Distillation and Behavior Foundation Models

Policy distillation trains a student policy from teacher behavior and has been used to compress, transfer, or combine learned policies [18]. Dataset aggregation and kickstarting-style methods additionally let the learner collect states and query an expert or teacher policy on the induced state distribution [19]. TacBFP follows this online teacher-labeling perspective, but the student is not merely a smaller policy. It is a behavior foundation model with a learned prior and frozen decoder, designed so that downstream reinforcement learning, imitation learning, and perception-conditioned control can reuse the same tactile motor substrate.

## III. Preliminaries

### A. Dexterous Reorientation

We consider a dexterous robotic hand manipulating a rigid object in simulation. At each control step, the environment state includes the hand joint positions, previous control targets, tactile contacts, object pose, and a target object orientation. The policy outputs a 22-dimensional hand action $\mathbf{a} \in \mathbb{R}^{22}$, which is converted to low-level position targets and applied through the simulator. The task reward encourages the object orientation to match the target while penalizing unstable hand motion, excessive work, and object drops.

### B. Behavior Prior Learning

The goal of behavior prior learning is to replace raw action exploration with a structured latent action. Instead of learning a downstream policy $\pi(\mathbf{a}|\mathbf{o}, \mathbf{g})$ directly in the hand action space, we learn a decoder $\pi_\psi(\mathbf{a}|\mathbf{x}, \mathbf{z})$ and a prior $p_\theta(\mathbf{z}|\mathbf{x})$. Downstream policies then produce latent actions, while the frozen decoder maps these latent choices to hand commands. This setting is useful only if the latent space contains behaviors that remain stable under closed-loop rollout, which motivates distilling from already trained specialist policies rather than fitting offline action snapshots alone.

## IV. Method

### A. Overview

TacBFP learns a tactile-conditioned dexterous behavior prior in three stages. First, we train eight reorientation specialist policies for a sphere with scale factors from 0.3 to 1.0. Second, we distill these specialists into one generalist variational latent controller with an encoder, a learned prior, and a decoder. Third, we freeze the prior and decoder and train downstream policies that output residual latent actions for harder objects and tasks. The method is designed so that downstream learning inherits a contact-aware motor substrate while specifying task intent through small latent corrections.

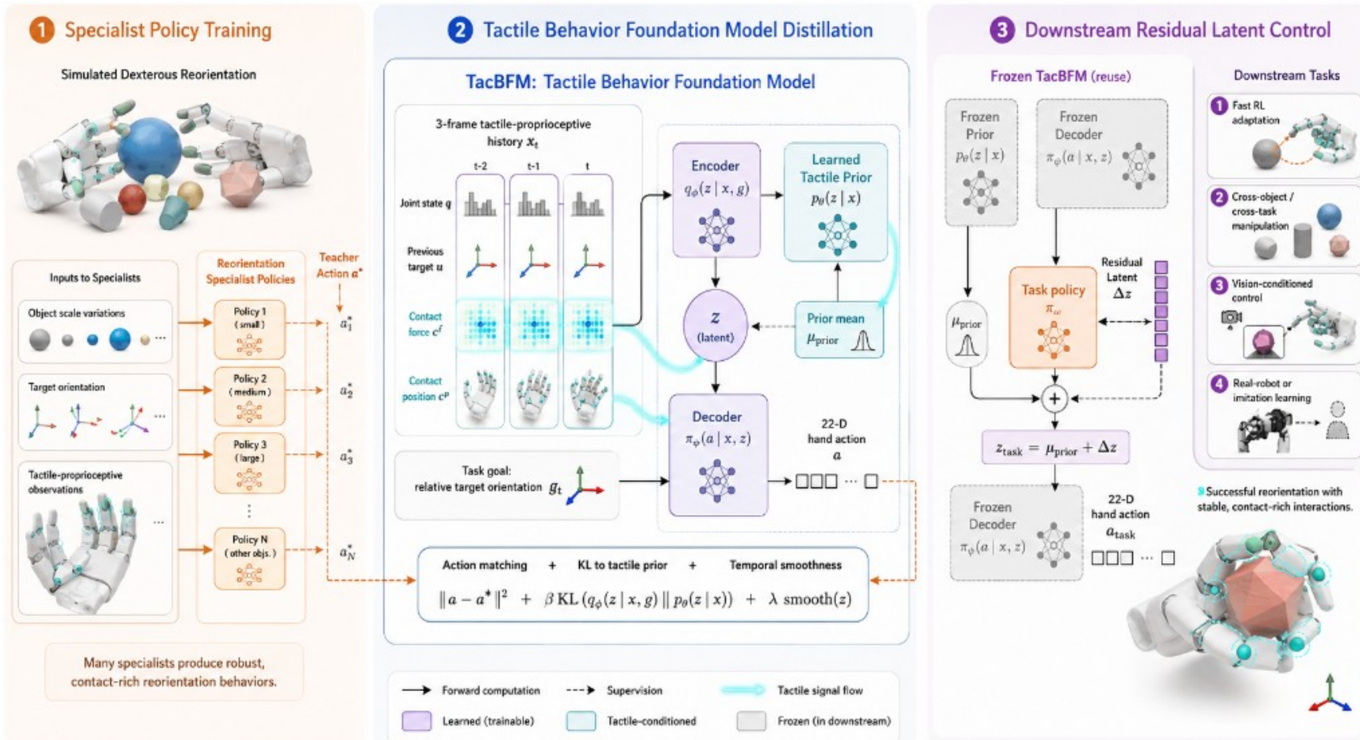


Fig. 1. Overview of TacBFP. The pipeline first trains sphere reorientation specialist policies over multiple scales, distills their tactile-proprioceptive closed-loop behavior into a generalist latent controller with a learned tactile prior, and finally reuses the frozen prior and decoder for downstream residual latent control on more complex objects and manipulation tasks.

### *B. Specialist Policy Training*

We train specialist policies for sphere reorientation using proximal policy optimization [1] in Isaac Gym/Isaac Lab-style massively parallel simulation [2]. The nominal sphere diameter is $8\,\mathrm{cm}$; the eight specialist policies use scale factors $\{0.3, 0.4, \ldots, 1.0\}$. Each specialist observes the teacher-compatible state $\mathbf{o}_t$, which consists of the tactile-proprioceptive history and task goal information, together with privileged object information used only by the teacher policy. The scale sweep exposes the hand to different aperture sizes and contact distributions while preserving a simple object geometry, producing a clean source of reusable contact behavior.

The specialist stage supplies the behavior coverage for the prior. Rather than asking a latent model to discover dexterous manipulation from scratch, we use trained specialists as teachers that can label the student's own rollouts. In the generalist distillation pipeline, environments are assigned to the eight sphere-scale teachers in a strided batch layout, each teacher produces the corresponding low-level target action, and the targets are interleaved back into a single training batch. This online multi-teacher setting keeps the supervised target aligned with the distribution induced by the current student while preserving scale-specific expert behavior.

### *C. Tactile-Conditioned State Representation*

The student state $\mathbf{x}_t$ is a compact tactile-proprioceptive history. In the current implementation, each frame contains normalized hand joint positions, previous control targets, five smoothed tactile contact magnitudes, and five 3D tactile contact positions. The model stacks the most recent three frames, yielding a 192-dimensional state:

$$\mathbf{x}_t = \left[\mathbf{q}_{t-2:t}, \mathbf{u}_{t-2:t}, \mathbf{c}^f_{t-2:t}, \mathbf{c}^p_{t-2:t}\right], \tag{1}$$

where $\mathbf{q}$ denotes hand joint state, $\mathbf{u}$ denotes previous control targets, $\mathbf{c}^f \in \mathbb{R}^5$ denotes per-finger contact magnitude, and $\mathbf{c}^p \in \mathbb{R}^{15}$ denotes per-finger 3D contact locations. The task observation $\mathbf{g}_t$ is the relative target orientation in the no-position downstream setting.

This representation is processed by a direct-concatenation MLP. The stacked joint state, previous control targets, tactile contact magnitudes, 3D contact locations, and object longest-side scale cue are concatenated into one vector and passed through fully connected layers. The resulting tactile-proprioceptive feature is shared by the encoder, prior, and decoder. This simple MLP design keeps the behavior model compact while still letting the latent prior react to changes in the current contact pattern and object scale.

This representation is the main difference between TacBFP and a purely kinematic behavior prior. The learned prior $p_\theta(\mathbf{z}|\mathbf{x}_t)$ does not merely predict what hand motion is likely for the current joint pose; it predicts what latent behavior is appropriate under the current tactile contact pattern and object scale. This allows the same object pose goal to induce different latent distributions when the hand is securely pinching, barely touching, or losing contact.

### D. Specialist-to-Generalist Variational Distillation

TacBFP uses an encoder-prior-decoder architecture for tactile behavior distillation. The encoder models the posterior distribution over latent actions conditioned on the tactile-proprioceptive feature and task goal:

$$q_\phi(\mathbf{z}_t|\mathbf{x}_t, \mathbf{g}_t) = \mathcal{N}\left(\boldsymbol{\mu}_t^{\mathrm{enc}}, \boldsymbol{\sigma}_t^{\mathrm{enc}}\right). \tag{2}$$

The learned prior predicts a state-conditioned latent distribution:

$$p_\theta(\mathbf{z}_t|\mathbf{x}_t) = \mathcal{N}\left(\boldsymbol{\mu}_t^{\mathrm{prior}}, \boldsymbol{\sigma}_t^{\mathrm{prior}}\right). \tag{3}$$

The decoder maps the tactile-proprioceptive feature and sampled latent code to a hand action:

$$\mathbf{a}_t = \pi_\psi(\mathbf{x}_t, \mathbf{z}_t). \tag{4}$$

The encoder, prior, and decoder operate on the concatenated tactile-proprioceptive MLP feature and use a 32-dimensional latent space.

During online distillation, the student rolls out in the environment and the assigned sphere-scale specialist provides the teacher action $\mathbf{a}_t^\star$ at the same state. The loss combines action matching, latent regularization to the learned prior, and temporal smoothness over consecutive posterior means:

$$\begin{aligned}\mathcal{L} = \; & \|\pi_\psi(\mathbf{x}_t, \mathbf{z}_t) - \mathbf{a}_t^\star\|_2^2 + \beta D_{\mathrm{KL}}\left(q_\phi(\mathbf{z}_t|\mathbf{x}_t, \mathbf{g}_t) \| p_\theta(\mathbf{z}_t|\mathbf{x}_t)\right) \\ & + \lambda \left\|\boldsymbol{\mu}_t^{\mathrm{enc}} - \rho \boldsymbol{\mu}_{t-1}^{\mathrm{enc}}\right\|_2. \end{aligned} \tag{5}$$

The KL term teaches the prior to predict useful latent codes without the task goal, while the temporal term discourages discontinuous latent jumps. The current generalist configuration uses online DAgger-style action distillation from the eight sphere-scale teachers, a KL latent regularizer, and a direct-concatenation MLP state encoder. In this configuration, $\beta$ is annealed from $0.01$ to $0.001$, $\lambda = 0.005$, $\rho = 0.99$, and $\mathbf{z} \in \mathbb{R}^{32}$.

### E. Residual Latent Downstream Control

After distillation, downstream tasks freeze the tactile state normalizer, the prior network, and the decoder. A downstream policy receives the normalized tactile-proprioceptive state and task observation, and predicts a residual latent action $\Delta\mathbf{z}_t$. The executed latent code is

$$\mathbf{z}_t^{\mathrm{task}} = \boldsymbol{\mu}_t^{\mathrm{prior}} + \Delta\mathbf{z}_t, \tag{6}$$

and the environment action is

$$\mathbf{a}_t^{\mathrm{task}} = \pi_\psi(\mathbf{x}_t, \mathbf{z}_t^{\mathrm{task}}). \tag{7}$$

This residual formulation keeps exploration near the tactile-conditioned prior while still allowing the task policy to move outside the default behavior when reward demands it. It also decouples reusable dexterous behavior from task-specific perception: visual features, demonstration encoders, or real-robot state estimators can be attached to the high-level policy while the same frozen prior and decoder produce hand actions.

## V. Experiments

The experiments test whether contact behavior distilled from simple sphere specialists can become a reusable tactile motor prior. We organize the evidence around five reviewer-facing questions:

- **Q1. Prior coverage and extrapolation:** Does the generalist prior preserve the sphere-scale teacher behaviors, and does it extrapolate to held-out scales and shapes?
- **Q2. Downstream transfer:** When the prior and decoder are frozen, does residual latent control improve reorientation on anisotropic objects?
- **Q3. Rotation reuse:** Can the same latent motor substrate support continuous rotation around a commanded axis?
- **Q4. Mechanism:** Which tactile, scale, and prior-design choices account for the improvement?
- **Q5. Embodiment transfer:** Does the hand prior remain useful when hand control is embedded inside an arm-hand policy?

### A. Experimental Setup

**Simulation and task setting.** All simulation evaluations use the same dexterous hand model, task reward family, object initialization protocol, and termination rules across methods within each table. The sphere-specialist teacher set uses scale factors $\{0.3, 0.4, \ldots, 1.0\}$ of the nominal $8\,\mathrm{cm}$ sphere. Held-out sphere scales 1.1 and 1.2 and the additional held-out object rows are never used as sphere-scale teachers, so they test extrapolation rather than teacher memorization.

**Evaluation protocol.** Unless otherwise stated, each reported row is evaluated over 1000 attempts with randomized evaluation seeds. Reorientation methods use the same target-orientation distribution and timeout cap within each benchmark. Rotation experiments use a 20-second window and report the signed accumulated rotation around the commanded $z$ axis. Training-curve figures report episode reward and episode length over agent steps to show both task progress and contact stability during learning.

**Baselines.** The main comparisons isolate three sources of improvement. *RL from scratch* learns the task directly without the distilled behavior prior. *w/o tactile* keeps the same generalist-prior pipeline but removes tactile sensing from the prior input. TacBFP uses the tactile-proprioceptive prior and frozen decoder, with downstream policies acting through residual latent codes. Downstream transfer tables additionally include actor-level transfer, action-space residual learning, single-scale priors, and decoder finetuning variants when those evaluations are available.

**Metrics and ranking.** We follow prior dexterous reorientation and in-hand rotation work by reporting success, drop/fallen rate, episode length, and rotation progress [11]–[13], [20]. Let $q_t$ and $q^g$ denote the object and goal quaternions, and let $p_t$ and $p^g$ denote the object and goal positions. The orientation error is the quaternion geodesic distance

$$e_R(t) = 2\sin^{-1}\left(\min\left(1, \left\|\mathrm{vec}\left(q_t \otimes (q^g)^{-1}\right)\right\|_2\right)\right). \tag{8}$$

A reorientation trial is successful if $\min_t e_R(t) \le 0.2$ rad before timeout or object drop. Object drop is defined by

TABLE I

SPHERE-SCALE AND UNSEEN-OBJECT EVALUATION OF THE GENERALIST PRIOR ABLATIONS. SUCCESS RATES ARE REPORTED WITH 95% CONFIDENCE INTERVALS. STEPS† COUNTS FAILED ATTEMPTS AS 100 STEPS. SCALES 1.1 AND 1.2 ARE UNSEEN SPHERE SCALES, AND THE ADDITIONAL OBJECT ROWS EVALUATE HELD-OUT OBJECT SHAPES OUTSIDE THE TEACHER SET.

| Case | From scratch | | w/o tactile | | Ours | |
|---|---|---|---|---|---|---|
| | Succ. ↑ | Steps† ↓ | Succ. ↑ | Steps† ↓ | Succ. ↑ | Steps† ↓ |
| 0.3 | 24.50±2.67% | 79.40±2.27 | 92.90±1.59% | 35.03±1.59 | **97.30±1.01%** | **31.07±1.26** |
| 0.4 | 25.20±2.69% | 79.81±2.21 | 96.10±1.20% | 29.76±1.16 | **98.40±0.78%** | **27.58±0.94** |
| 0.5 | 99.80±0.28% | **3.28±0.30** | **100.00±0.00%** | 3.42±0.18 | 99.90±0.20% | 3.45±0.27 |
| 0.6 | 20.40±2.50% | 84.18±2.00 | 96.50±1.14% | 33.33±1.26 | **98.90±0.65%** | **29.72±0.98** |
| 0.7 | 14.60±2.19% | 89.27±1.67 | 94.20±1.45% | 38.31±1.43 | **97.00±1.06%** | **35.29±1.11** |
| 0.8 | 10.20±1.88% | 92.95±1.47 | 92.90±1.59% | 41.73±1.36 | **96.80±1.09%** | **39.34±1.17** |
| 0.9 | 8.40±1.72% | 93.95±1.30 | 91.40±1.74% | 50.63±1.50 | **92.00±1.68%** | **49.81±1.50** |
| 1.0 | 5.10±1.36% | 96.11±1.05 | **85.40±2.19%** | 61.13±1.66 | 84.70±2.23% | **60.88±1.66** |
| **Seen Avg.** | 26.03±21.29% | 77.37±21.21 | 93.67±2.97% | 36.67±11.68 | **95.62±3.47%** | **34.64±11.72** |
| 1.1 | 4.90±1.34% | 96.20±1.05 | 55.50±3.08% | 85.10±1.39 | **62.00±3.01%** | **83.10±1.49** |
| 1.2 | 3.30±1.11% | **97.52±0.84** | 1.70±0.80% | 100.04±0.44 | **13.30±2.11%** | 99.80±0.98 |
| block_01 | 18.30±2.40% | 85.26±1.95 | 58.20±3.06% | 72.56±2.07 | **79.50±2.50%** | **61.10±1.92** |
| trash_can_01 | 11.70±1.99% | 90.62±1.61 | 14.60±2.19% | 94.56±1.25 | **79.50±0.00%** | **61.10±0.00** |
| switch_01 | 13.80±2.14% | 89.30±1.68 | 28.70±2.81% | 85.27±1.72 | **47.10±3.10%** | **80.49±1.90** |
| box_01 | 13.50±2.12% | 89.69±1.65 | 29.00±2.81% | 90.24±1.68 | **41.80±3.06%** | **88.24±1.99** |
| bottle_03 | 11.80±2.00% | 90.55±1.62 | 57.70±3.06% | 69.21±1.91 | **77.80±2.58%** | **59.99±1.79** |
| strawberry | 14.50±2.18% | 88.40±1.76 | 75.00±2.69% | 52.26±1.87 | **82.00±2.38%** | **47.83±1.69** |
| **Unseen Avg.** | 11.47±3.47% | 90.94±2.80 | 40.05±17.51% | 81.16±10.81 | **60.38±17.06%** | **72.71±12.27** |

$e_p(t) = \|p_t - p^g\|_2 \geq 0.03$ m or by a height-based reset. Steps† counts failed attempts at the episode cap, so lower values reward both faster success and fewer failures. All table entries use mean ± 95% confidence intervals unless otherwise stated; highlighting marks the first, second, and third ranks by the mean under each metric direction.

### B. Generalist Prior Coverage and Held-Out Extrapolation

We first evaluate the specialist-to-generalist training stage and the boundary of its extrapolation. The student is trained with online multi-teacher distillation and the direct-concatenation MLP state encoder described in Sec. III. Table I reports seen sphere scales, held-out sphere scales, and held-out object shapes using the same success and capped-steps metrics. This test asks whether one tactile prior and decoder can retain the teacher family while remaining useful outside the teacher distribution.

### C. Residual Latent Transfer to Complex Reorientation

We next freeze the generalist prior and decoder and train downstream residual policies on harder anisotropic objects. These objects differ from the sphere teachers in geometry, size, and contact affordances, so the experiment tests whether sphere-trained contact behavior can serve as a reusable substrate rather than a sphere-specific policy. Table II compares raw-action PPO, actor-level transfer, action-space residual learning, single-scale priors, and TacBFP variants under the same task reward, environment count, training budget, object initialization, and evaluation protocol. Figure 2 shows the corresponding training curves.

Table II shows that the strongest variants are the two TacBFP residual-latent policies. Finetuning the decoder gives the best success and capped-step scores on block_03, block_05, bottle_01, and trash_can_02, while the frozen-decoder variant gives the best multi-object success. Action-space residual learning performs well on fruit_02 but degrades sharply in the Multi setting, suggesting that residuals in the low-level action space do not provide the same reusable contact structure across objects. The reward and episode-length curves in Fig. 2 support the same interpretation: TacBFP variants rise quickly and maintain longer episodes, whereas actor-level transfer and single-scale priors plateau lower, especially in the Multi condition. The episode-length curves are important because higher reward coincides with longer controlled interaction, indicating that the gains come from maintaining stable contact rather than only exploiting short successful bursts.

### D. Continuous $z$-Axis Rotation Reuse

The previous experiments evaluate goal-conditioned reorientation. We also test whether the same latent motor substrate can be reused for continuous rotation around the vertical axis, where the policy must keep contact while accumulating signed rotation instead of stopping at a target pose. Table III reports the current $z$-axis rotation results; remaining baseline entries are left blank until their matching evaluations are available.

### E. Mechanism and Design Ablations

We isolate the design choices that make the prior useful. The planned ablations remove tactile contact forces, remove 3D contact positions, remove tactile inputs from the MLP state encoder, remove object longest-side conditioning, remove the learned prior mean in downstream control, and add auxiliary privileged object-state prediction heads. This table is intended to connect each architectural claim to a controlled variant once the corresponding evaluations are complete.

### F. Arm-Hand Embodiment Transfer

The final evaluation embeds the frozen hand prior into tasks that also require a robot arm. In this setting, the arm positions the hand relative to the object or workspace, while the hand controller acts through TacBFP's latent residual interface. This experiment asks whether the hand prior remains useful when an upstream arm policy changes approach direction, gravity loading, and reachable workspace constraints. We compare against an arm-hand policy trained entirely in the raw action space and a modular baseline that uses the same arm controller but no tactile hand prior.

## VI. DISCUSSION

TacBFP is best understood as a behavior foundation model rather than a single-task controller. The pretrained prior captures what contact-rich hand behavior is plausible from the current tactile-proprioceptive state, while downstream policies specify which behavior should be selected for the current task. This separation is useful because it makes downstream learning less dependent on random low-level exploration and more compatible with heterogeneous task inputs.

Several implementation-guided extensions are natural next steps. First, the downstream residual policy can share its hidden features with auxiliary prediction heads for object position, object orientation, and object angular velocity. In simulation these targets are privileged labels, so the auxiliary

TABLE II
DOWNSTREAM REORIENTATION ON COMPLEX OBJECTS. THE PRIOR AND DECODER ARE FROZEN FOR TACBFP; ONLY THE RESIDUAL LATENT POLICY IS TRAINED. MULTI JOINTLY EVALUATES BLOCK_02, BLOCK_06, FRUIT_01, BOTTLE_02, AND SPHERE_02.

| | block_03 | | block_05 | | bottle_01 | | fruit_02 | | trash_can_02 | | Multi | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | Succ. ↑ | Steps ↓ | Succ. ↑ | Steps ↓ | Succ. ↑ | Steps ↓ | Succ. ↑ | Steps ↓ | Succ. ↑ | Steps ↓ | Succ. ↑ | Steps ↓ |
| RL from scratch | 1.40±0.73% | 98.73±0.66 | 2.10±0.89% | 98.09±0.81 | 1.70±0.80% | 98.42±0.74 | 0.80±0.55% | 99.26±0.52 | 0.90±0.59% | 99.17±0.54 | 1.00±0.62% | 99.08±0.57 |
| RL with pretrained actor | 70.40±2.83% | 42.43±2.34 | 54.60±3.09% | 54.33±2.59 | 60.70±3.03% | 50.97±2.47 | 73.00±2.75% | 43.13±2.16 | 44.10±3.08% | 63.77±2.55 | 40.00±3.04% | 69.76±2.33 |
| Action-space residual | 73.20±2.75% | 42.98±2.17 | 52.20±3.10% | 57.37±2.54 | 60.40±3.03% | 54.03±2.34 | **82.40±2.36%** | **36.41±1.85** | 47.80±3.10% | 64.67±2.33 | 8.80±1.76% | 93.54±1.30 |
| Single-scale prior | 73.70±2.73% | 41.10±2.21 | 59.50±3.04% | 50.02±2.57 | 65.20±2.95% | 48.01±2.38 | 72.20±2.78% | 44.16±2.17 | 67.20±2.91% | 46.45±2.35 | 65.80±2.94% | 44.27±2.51 |
| TacBFP w. frozen decoder | 72.80±2.76% | 40.65±2.27 | 61.40±3.02% | 48.64±2.54 | 67.40±2.91% | 46.57±2.33 | 72.70±2.76% | 43.99±2.15 | 72.00±2.78% | 43.75±2.20 | **73.30±2.74%** | 41.69±2.20 |
| TacBFP w. finetuning decoder | **78.60±2.54%** | **35.76±2.11** | **68.10±2.89%** | **42.47±2.45** | **70.10±2.84%** | **43.77±2.30** | 75.20±2.68% | 41.32±2.11 | **73.80±2.73%** | **43.19±2.14** | 70.00±2.84% | **39.23±2.48** |

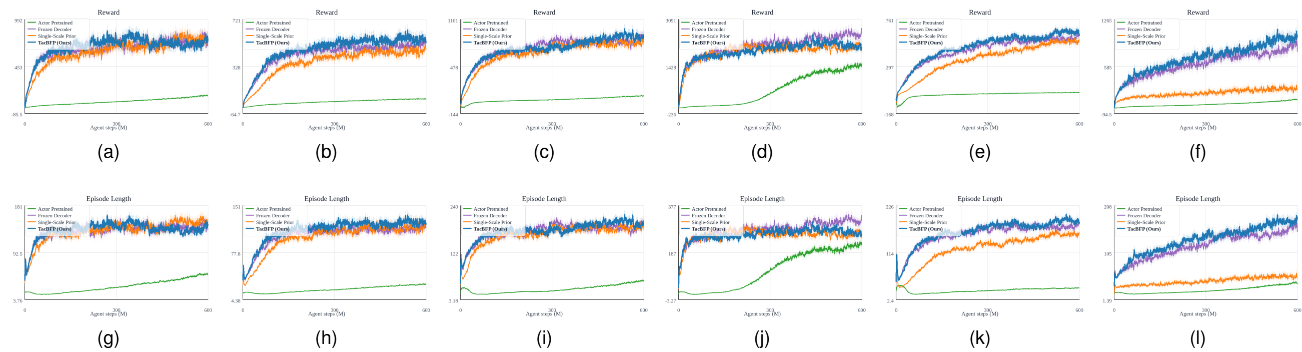


Fig. 2. Downstream repose training curves for the complex-object transfer ablations. The columns, from left to right, correspond to block_03, block_05, bottle_01, fruit_02, trash_can_02, and Multi. Subfigures (a)–(f) show episode reward, and (g)–(l) show episode length. Curves are plotted up to 600M agent steps; shaded regions indicate rolling 95% confidence intervals.

TABLE III
DOWNSTREAM ROTATION AROUND THE $z$ AXIS. ROTATION ANGLES ARE REPORTED WITH 95% CONFIDENCE INTERVALS OVER 1000 ATTEMPTS. BASELINE RESULTS ARE LEFT BLANK PENDING EVALUATION.

| Method | block_01 | block_03 | bottle_01 | fruit_02 | switch_01 | trash_can_02 |
|---|---|---|---|---|---|---|
| From scratch w/o tactile | 93.89±3.16 | 50.55±2.69 | **735.26±16.90** | **843.98±21.31** | 834.19±15.56 | **194.30±9.92** |
| TacBFP | **109.20±4.45** | **342.39±19.56** | 56.67±2.37 | 35.09±2.90 | | 56.15±2.81 |

TABLE IV
ABLATIONS ON DOWNSTREAM REORIENTATION. EACH ROW CHANGES ONE COMPONENT OF TACBFP WHILE KEEPING THE SAME DOWNSTREAM TRAINING PROTOCOL.

| Variant | Success ↑ | Fallen ↓ | Steps ↓ | Reward ↑ |
|---|---|---|---|---|
| TacBFP | | | | |
| w/o contact force | | | | |
| w/o contact position | | | | |
| w/ tactile-proprioceptive concat MLP | | | | |
| w/ proprioceptive-only MLP | | | | |
| w/o object scale cue | | | | |
| w/o prior mean | | | | |
| with object-state head | | | | |

loss can improve state awareness without changing the deployment observation interface; at test time the heads are used only as representation shaping unless explicit state estimation is desired. Second, downstream learning can add a residual regularizer, such as $\|\Delta \mathbf{z}_t\|_2^2$ or a KL penalty around the tactile prior, to prevent the task policy from leaving the distilled behavior manifold too early. Third, the current model uses a three-frame concatenated history and diagonal Gaussian latent distribution; recurrent, temporal-convolutional, transformer, mixture-prior, or discrete-latent variants may better capture multimodal contact strategies such as pinch, roll, regrasp, and slip recovery. Finally, tactile inputs can be structured as per-finger tokens with contact dropout and force noise during training, which would make the tactile-conditioning claim stronger under sensor degradation.

TABLE V
ARM-HAND MANIPULATION TRANSFER. THE TABLE MEASURES WHETHER THE FROZEN HAND PRIOR REMAINS USEFUL WHEN MANIPULATION ALSO REQUIRES ARM MOTION.

| Method | Task Success ↑ | Drop Rate ↓ | Steps ↓ | Reward ↑ |
|---|---|---|---|---|
| Raw arm-hand PPO | | | | |
| Arm + hand latent decoder | | | | |
| Arm + TacBFP | | | | |

The evaluation also clarifies the boundaries of the approach. The prior is trained from sphere-scale specialists, so transfer to complex objects depends on whether their manipulation can reuse the same contact primitives. Highly concave, deformable, or slippery objects may require additional specialist coverage or online adaptation. The arm-hand setting introduces another boundary: the hand prior assumes a reasonable object presentation from the arm policy, so large approach errors can still break contact before the prior becomes useful.

## VII. CONCLUSION

We presented TacBFP, a tactile behavior foundation model for dexterous in-hand reorientation. The method distills eight sphere-scale reorientation specialists into a generalist variational latent controller with a learned tactile-proprioceptive prior and a frozen decoder. Downstream policies act through residual latent codes added to the prior mean, enabling stable contact behaviors learned from simple spheres to be reused for harder anisotropic objects and arm-hand manipulation tasks.

# References


[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[2] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac gym: High performance gpu-based physics simulation for robot learning," *arXiv preprint arXiv:2108.10470*, 2021.

[3] Z. Luo, J. Cao, J. Merel, A. Winkler, J. Huang, K. Kitani, and W. Xu, "Universal humanoid motion representations for physics-based control," in *International Conference on Learning Representations*, 2024.

[4] X. B. Peng, Y. Guo, L. Halper, S. Levine, and S. Fidler, "ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters," *arXiv preprint arXiv:2205.01906*, 2022.

[5] C. Tessler, Y. Kasten, Y. Guo, S. Mannor, G. Chechik, and X. B. Peng, "CALM: Conditional adversarial latent models for directable virtual characters," in *ACM SIGGRAPH Conference Proceedings*, 2023.

[6] J. Merel, L. Hasenclever, A. Galashov, A. Ahuja, V. Pham, G. Wayne, Y. W. Teh, and N. Heess, "Neural probabilistic motor primitives for humanoid control," *arXiv preprint arXiv:1811.11711*, 2018.

[7] H. Yao, Z. Song, B. Chen, and L. Liu, "ControlVAE: Model-based learning of generative controllers for physics-based characters," *arXiv preprint arXiv:2210.06063*, 2022.

[8] J. Won, D. Gopinath, and J. Hodgins, "Physics-based character controllers using conditional VAEs," *ACM Transactions on Graphics*, vol. 41, no. 4, pp. 1–12, 2022.

[9] Q. Zhu, H. Zhang, M. Lan, and L. Han, "Neural categorical priors for physics-based character control," *arXiv preprint arXiv:2308.07200*, 2023.

[10] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, and S. Levine, "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations," in *Proceedings of Robotics: Science and Systems*, 2018.

[11] T. Chen, J. Xu, and P. Agrawal, "A system for general in-hand object re-orientation," in *Proceedings of the 5th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 164, 2022, pp. 297–307. [Online]. Available: https://proceedings.mlr.press/v164/chen22a.html

[12] H. Qi, A. Kumar, R. Calandra, Y. Ma, and J. Malik, "In-hand object rotation via rapid motor adaptation," in *Conference on Robot Learning*, 2022.

[13] H. Qi, B. Yi, S. Suresh, M. Lambeta, Y. Ma, R. Calandra, and J. Malik, "General in-hand object rotation with vision and touch," in *Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 2549–2564. [Online]. Available: https://mlanthology.org/corl/2023/qi2023corl-general/

[14] Z.-H. Yin, B. Huang, Y. Qin, Q. Chen, and X. Wang, "Rotating without seeing: Towards in-hand dexterity through touch," in *Proceedings of Robotics: Science and Systems*, 2023.

[15] Y. Yuan, H. Che, Y. Qin, B. Huang, Z.-H. Yin, K.-W. Lee, Y. Wu, S.-C. Lim, and X. Wang, "Robot synesthesia: In-hand manipulation with visuotactile sensing," in *IEEE International Conference on Robotics and Automation*, 2024, pp. 6558–6565.

[16] I. Guzey, B. Evans, S. Chintala, and L. Pinto, "Dexterity from touch: Self-supervised pre-training of tactile representations with robotic play," in *Proceedings of the 7th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 3142–3166. [Online]. Available: https://proceedings.mlr.press/v229/guzey23a.html

[17] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, "3D-ViTac: Learning fine-grained manipulation with visuo-tactile sensing," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2024, pp. 2557–2578. [Online]. Available: https://mlanthology.org/corl/2024/huang2024corl-3dvitac/

[18] A. A. Rusu, S. G. Colmenarejo, C. Gulcehre, G. Desjardins, J. Kirkpatrick, R. Pascanu, V. Mnih, K. Kavukcuoglu, and R. Hadsell, "Policy distillation," in *International Conference on Learning Representations Workshop*, 2016.

[19] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," *arXiv preprint arXiv:1011.0686*, 2010.

[20] OpenAI, M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, J. Schneider, S. Sidor, J. Tobin, P. Welinder, L. Weng, and W. Zaremba, "Learning dexterous in-hand manipulation," *The International Journal of Robotics Research*, vol. 39, no. 1, pp. 3–20, 2020.